

Souvik De

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TECHNOLOGIES

LANGUAGES
JavaScript, TypeScript, Go, Python, SQL

BACKEND
Node.js, FastAPI, Gin

AI SYSTEMS
RAG Architectures, Vector Database(Qdrant), LLM Integration, Langchain, Ollama, Ragas, DeepAgents, Claude, OpenAI, DeepEval, Langfuse, Mineru

DATABASES
MySQL, Firebase, Postgres, Redis

DEVOPS
Docker, GitHub Actions, CI/CD

EDUCATION

Chandigarh University
BE in Computer Science
June 2018 – June 2022

- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Software Engineering, System Design

Backend Software Engineer with 4+ years of experience building production AI infrastructure, distributed backend systems, and developer tooling. Experienced in designing RAG platforms, document ingestion pipelines, vector search systems, and LLM orchestration using Go, TypeScript, and Python. Former Postman engineer, AsyncAPI maintainer, and Google Summer of Code mentor with experience building open-source tools used by thousands of developers.

EXPERIENCE

Software Engineer II, XANE.AI – Gurgaon July 2025 – Present

- Built a multi-stage OCR → chunking → embedding → retrieval pipeline powering Maruti Suzuki's AI assistant, replacing manual data entry and cutting content-preparation time to enable semantic search across thousands of vehicle manuals.
- Architected an OCR-to-RAG ingestion flow converting unstructured PDFs into structured embeddings indexed in a vector database, giving AI agents persistent knowledge memory.
- Introduced a RAG evaluation framework on Ragas that surfaced an answer-relevance gap and drove it from 64% to 89%, accelerating the team's iteration cycle on retrieval quality.

Lead Software Engineer, CodeMate.AI – Noida December 2024 – May 2025

- Engineered an enhanced search algorithm for Swagger files, improving search precision by 10-20% and RAG system performance.
- Set up a CI/CD pipeline with GitHub Actions to automate release cycles across macOS, Linux, and Windows, making cross-platform deployments faster and more reliable.
- Optimized a local server used by our VS Code extension, cutting startup time by over 30% and improving the developer experience.
- Automated deployment from GitHub Actions to a production VM and configured branch protection rules, improving code quality and release stability while saving the team 2-5 hours per week of manual release work.

Software Engineer, Postman – Bangalore Feb 2022 – June 2024

- Built and maintained a cross-platform CLI integrating official AsyncAPI tooling (20,000+ downloads), giving developers a unified workflow.
- Shipped AsyncAPI Bundler (16,000+ downloads), resolving complex json ref dependency graphs spanning specification files.
- Served as maintainer across 3+ open-source projects, reviewing 185+ PRs, promoting two contributors to maintainers, and guiding architectural roadmap decisions.
- Mentored 3-5 contributors through Google Summer of Code (GSoC), taking their projects from proposal to production-ready completion.
- Co-led governance initiative to standardize parser tooling architecture across multiple languages, improving maintainability and reducing duplication.

Backend Developer Intern, Mage – Remote Nov 2021 – Feb 2022

- Built microservices in Go to automate cloud infrastructure provisioning using templated configuration systems.
- Developed a React-based low-code interface for generating and provisioning infrastructure on demand.
- Deployed containerized services using Docker and Kubernetes in cloud-native environments.

PROJECTS

Kitsu

[github](#)

AI-powered GitHub Action for automated PR review summaries

- Built a GitHub Action that analyzes git patches using LLMs to generate concise pull request summaries.
- Handles large patch files by filtering low-signal changes and chunking oversized diffs.
- Designed to run entirely inside GitHub Actions with minimal latency.

Notion-Board

[github](#)

GitHub Action to sync GitHub Issues to Notion Database

- Developed a GitHub Action to auto sync and update GitHub issues and its state with Notion Pages.
- Earned recognition from the open-source community through project stars and valuable user feedback.
- Tools Used: TypeScript, GitHub Actions.

AsyncAPI-RAG

[github](#)

FastAPI application that scrapes AsyncAPI documentation to create a RAG system and provides a chat API to answer user queries.

- Built data scraper to scrape AsyncAPI spec and tools documentation from GitHub.
- Used LangChain to chunk Markdown data, generated embeddings with the Nomic embed model from Ollama, stored embeddings in a Qdrant database, and exposed a REST API to handle user queries and return contextual answers.
- Tools Used: Python, qdrant, Langchain, ollama, docker